

Durwas Khot

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SUMMARY

Mechanical Design Engineer with strong hands-on experience designing hydraulic-mechanical industrial equipment from CAD concept through FEA validation and manufacturing specification. My background includes leading ISO 9001 internal audits, applying FMEA and 8D methodology to resolve root-cause production issues, and collaborating cross-functionally to reduce manufacturing cost and complexity. Seeking a role where I can bring rigorous design-for-manufacture thinking to the development of special-purpose vehicles and equipment.

EDUCATION

MSc Automotive Engineering with EVs

Oxford Brookes University

Distinction (1:1)

September 2024 - September 2025

BE - Mechanical Engineering

University of Mumbai

77.66% (UK Equivalent – 1:1)

August 2019 - May 2023

WORK EXPERIENCE

Junior Design Engineer, TRITORC Equipments

June 2023 – August 2024

- Designed and developed industrial mechanical systems, including hydraulic machines, generating 3D CAD models, 2D manufacturing drawings, and FEA to validate structural strength, load distribution, and operational reliability. Supported in manufacturing problem resolution through root cause analysis.
- Maintained comprehensive technical documentation, CAD data, engineering change process control, and bill of materials (BOM), along with implementation of DFMEA and PFMEA for complex assemblies.
- Led collaboration with cross-functional teams such as manufacturing, procurement, quality and service to improve machine build, prototype development, and reduce manufacturing complexity by iterative design reviews.

RESEARCH AND INDUSTRY PROJECTS

Oxford Brookes University (In collaboration with Bentley Motors, UK)

Master's Thesis Student

May 2025 – September 2025

- Conducted correlation of high-frequency (> 80 Hz) road-induced vibration in vehicle structure, through tests and flexible multi-body dynamics simulation to improve model prediction accuracy by 30%.
- Led model validation and correlation with physical test data, identifying key vibration drivers and delivering design insights that reduced steering wheel VTV from 2.38 to 1.64 m/s², validated against ISO 5349 comfort metrics.

Oxford Brookes University

September 2024 – March 2025

- Reduced cabin noise from 91.9 dB to 82.853 dB through ANSYS vibroacoustic analysis, identifying and optimising resonant frequencies. Improved ride comfort by strategically positioning the absorption materials at critical frequencies. Conducted vibroacoustic modelling of the vehicle architecture, with scope for multi-source noise integration and future experimental work.

Formula Student team - Modified Auto Club Racing, Thane (Founding member)

Team Manager and Suspension Lead

January 2022 – December 2022

Head - Vehicle Dynamics Department

May 2021 - December 2021

- Led a 30-member engineering team in the design, development, prototyping, manufacturing and assembly of an FS electric race car. Coordinated timelines, subsystem integration, and compliance with FS technical regulations.
- Led the development of a double-wishbone suspension system, performing MBD simulations in MSC Adams and CAD modelling in CATIA V5, optimising parameters including ± 40 mm wheel travel, and roll centre height of 35–45 mm to improve vehicle stability and performance, contributing to the team winning the Best Debut Team Award.

SKILLS

Software Tools: SolidWorks (Certified Solidworks Associate), CATIA, ANSYS, MSc Adams, MATLAB, Simulink/Simscape, Microsoft Office

Engineering: CAD/CAM/CAE, FEA, DFMEA, PMFEA, Multi Body Dynamics, GD&T, Technical Documentation

Management: Engineering Project Management, ISO 9001 Quality Management systems, ISO 14001 Environmental management systems, Lean Principles, 8D Methodology

Soft Skills: Leadership, Collaboration, Analytical thinking, Agility, Flexibility, Problem solving, Self-motivation